Very Low Noise, Fast Turn On, 50 mA Low Dropout Voltage Regulator

NCP508

The NCP508 is a 50 mA low noise voltage regulator, designed to exhibit fast turn on time and high ripple rejection. Each device contains a voltage reference unit, an error amplifier, a PMOS power transistor, resistors for setting output voltage, current limit, and temperature limit protection circuits.

The NCP508 has been designed for use with ceramic capacitors. The device is housed in SC-88A and WDFN6 1.5x1.5 packages. Standard voltage versions are 1.5, 1.8, 2.5, 2.8, 3.0, and 3.3. Other voltages are available in 100 mV steps.

Features

- Very Low Noise at 39 μVrms without a Bypass Capacitor
- High Ripple Rejection of 70 dB at 1 kHz
- Low Dropout Voltage of 140 mV (typ) at 30 mA
- Tight Load Regulation, typically 6 mV for ΔI_{out} = 50 mA
- Fast Enable Turn-On time of 20 μsec
- Logic Level Enable
- ESR can vary from a few mΩ to 3 Ω
- These are Pb-Free Devices

Typical Applications

- RF Subsystems in Handsets
- Noise Sensitive Circuits; VCOs, PLL

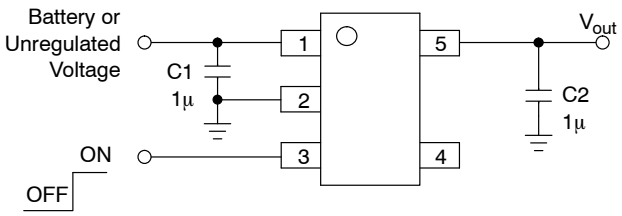
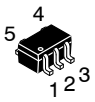
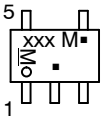


Figure 1. Typical Application Diagram



SC70-5/SC-88A/SOT-353
SQ SUFFIX
CASE 419A

MARKING DIAGRAM

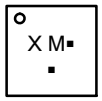


XXX = Specific Device Code
M = Date Code*
▪ = Pb-Free Package
(Note: Microdot may be in either location)

* Date Code orientation and/or position may vary depending upon manufacturing location.

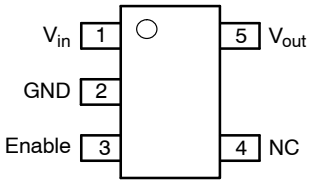


WDFN6
MN SUFFIX
CASE 511BJ

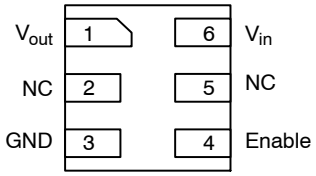


X = Specific Device Code
M = Date Code
▪ = Pb-Free Package
(Note: Microdot may be in either location)

PIN CONNECTIONS



SC-88A
(Top View)



WDFN6
(Top View)

ORDERING INFORMATION

See detailed ordering and shipping information on page 13 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 13.

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PIN FUNCTION DESCRIPTION

Pin No.	Pin Name	Description
1	V_{in}	Positive power supply input voltage
2	GND	Power supply ground
3	Enable	This input is used to place the device into low-power stand by. When this input is pulled low, the device is disabled. If this function is not used, Enable should be connected to V_{in}.
4	N/C	Not connected pin
5	V_{out}	Regulated output voltage

MAXIMUM RATING

Rating	Symbol	Value	Unit
Input Voltage	$V_{in(max)}$	13.0	V
Enable Voltage	Enable	−0.3 to $V_{in(max)}$ + 0.3	V
Output Voltage	V_{out}	−0.3 to $V_{in(max)}$ + 0.3	V
Power Dissipation and Thermal Characteristics (SC-88A) Power Dissipation Thermal Resistance, Junction-to-Ambient (Note 4)	 P_D $R_{\theta JA}$	 Internally Limited 200	 W °C/W
Power Dissipation and Thermal Characteristics (WDFN6) Power Dissipation Thermal Resistance, Junction-to-Ambient (Note 4)	 P_D $R_{\theta JA}$	 Internally Limited 313	 W °C/W
Maximum Junction Temperature	T_J	+125	°C
Operating Ambient Temperature	T_A	−40 to +85	°C
Storage Temperature	T_{stg}	−55 to +150	°C
Lead Soldering Temperature @ 260 °C	T_{solder}	10	sec

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. This device series contains ESD protection and exceeds the following tests:
   Human Body Model 2000 V per MIL-STD-883, Method 3015.
   Machine Model Method 200 V
2. Latch up Capability (85 °C) ± 100 mA DC with trigger voltage
3. Maximum package power dissipation limits must be observed.

$$P_D = \frac{T_{J(max)} - T_A}{R_{\theta JA}}$$

4. $R_{\theta JA}$ on a 30 x 30 mm PCB Cu thickness 1 oz; T_A = 25 °C.

RECOMMENDED OPERATING CONDITIONS

Rating	Symbol	Max	Unit
Maximum Operating Input Voltage	V_{in}	7.0	V

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

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ELECTRICAL CHARACTERISTICS (V_{in} = $V_{out(nom)}$ + 1.0 V, V_{enable} = V_{in}, C_{in} = 1.0 μF, C_{out} = 1.0 μF, T_J = 25 °C, unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit
Output Voltage Tolerance (T_A = 25 °C, I_{out} = 10 mA)	V_{out}	−2	–	+2	%
Output Voltage Tolerance (T_A = −40 °C to 85 °C, I_{out} = 10 mA)	V_{out}	−3	–	+3	%
Line Regulation (V_{in} = V_{out} + 1 V to 12 V, I_{out} = 10 mA) (Note 5)	Reg_{line}	–	2	20	mV
Load Regulation (I_{out} = 1.0 mA to 50 mA) (Note 5)	Reg_{load}	–	6	40	mV
Output Current (V_{out} = $V_{out(nom)}$ − 0.1 V)	$I_{out(nom)}$	50	–	–	mA
Dropout Voltage (V_{out} = 3.0 V, Measured at V_{out} − 100 mV) I_{out} = 30 mA I_{out} = 40 mA I_{out} = 50 mA	V_{in}–V_{out}	 – – –	 140 155 180	 250 300 –	mV
Quiescent Current (Enable Input = 0V)	I_Q	 –	 0.1	 1	μA
Ground Current (Enable Input = V_{in}, V_{in} = V_{out} + 1 V, I_{out} = 0 mA) (Enable Input = V_{in}, I_{out} = 1 mA) (Enable Input = V_{in}, I_{out} = 10 mA) (Enable Input = V_{in}, I_{out} = 50 mA)	I_{GND}	 – – – –	 145 160 300 1100	 200 260 500 1900	μA
Enable Input Threshold Voltage (Voltage Increasing, Output Turns On, Logic High) (Voltage Decreasing, Output Turns Off, Logic Low)	$V_{th(en)}$	 0.9 –	 – –	 – 0.15	V
Enable Input Current (V_{enable} = 2.4 V)	I_{enable}	–	8.0	15	μA
Output Turn On Time (Note 6)	–	–	20	–	μs
Output Short Circuit Current Limit (V_{out} = 0 V)	$I_{out(max)}$	100	250	–	mA
Ripple Rejection (V_{in} = $V_{out(nom)}$ + 1 Vdc + 0.5 V_{pp}, f = 1 kHz, Io = 10 mA)	RR	–	70	–	dB
Output Noise Voltage (f = 100 Hz to 100 kHz) (V_{out} = 1.5 V)	V_n	–	39	–	μVrms

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

5. Low duty cycle pulse techniques are used during testing to maintain the junction temperature as close to ambient as possible.
6. Turn on time is defined from Enable at 10% to V_{out} at 95% nominal value. Min and max values T_A = −40 °C to 85 °C, T_{jmax} = 125 °C. V_{enable} = 0 V to V_{in}. C_{out} = 1.0 μF.

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TYPICAL CHARACTERISTICS

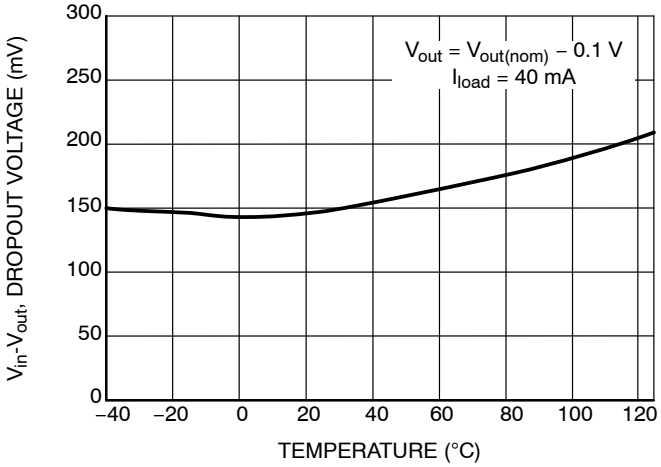


Figure 2. Dropout Voltage vs. Temperature, 1.5 V

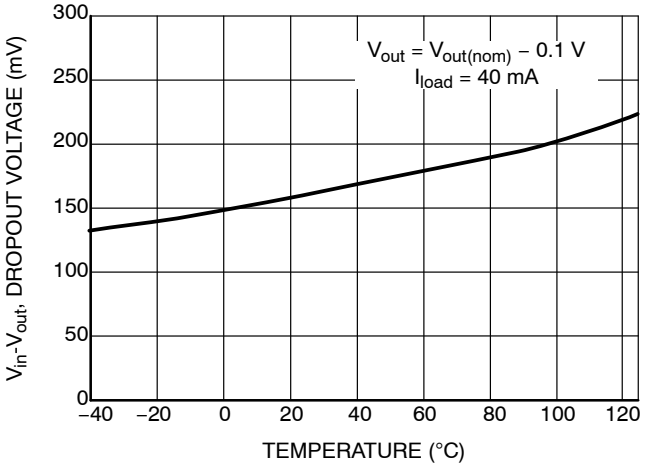


Figure 3. Dropout Voltage vs. Temperature, 3.3 V

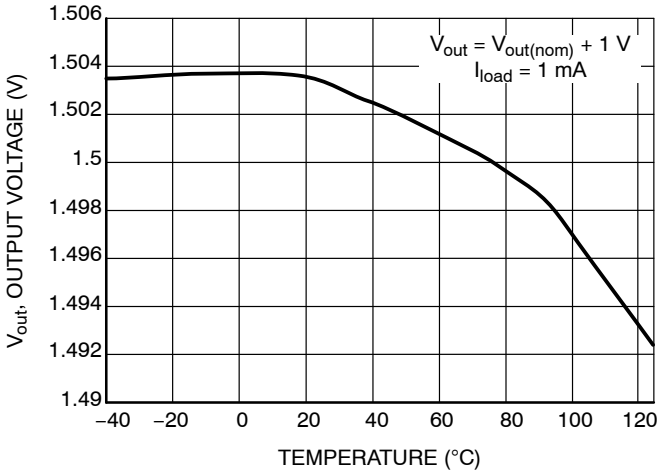


Figure 4. Output Voltage vs. Temperature, 1.5 V

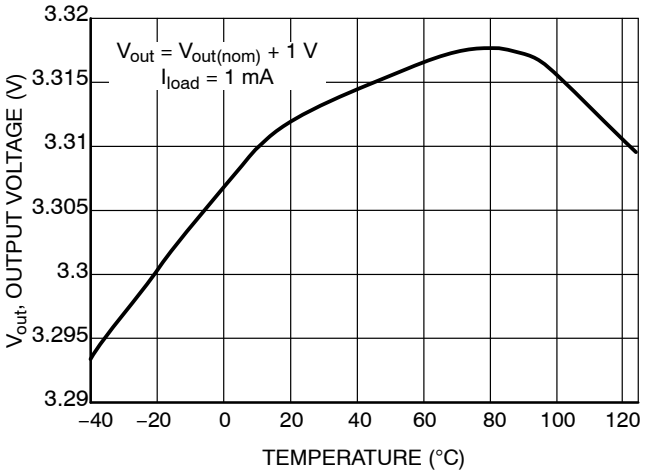


Figure 5. Output Voltage vs. Temperature, 3.3 V

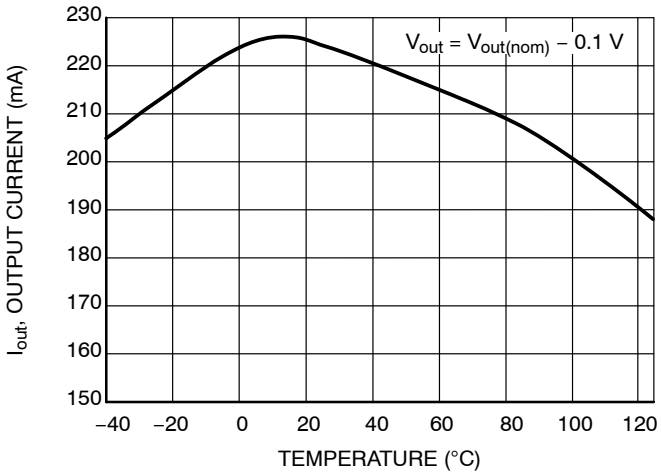


Figure 6. Output Current Limit vs. Temperature, 1.5 V

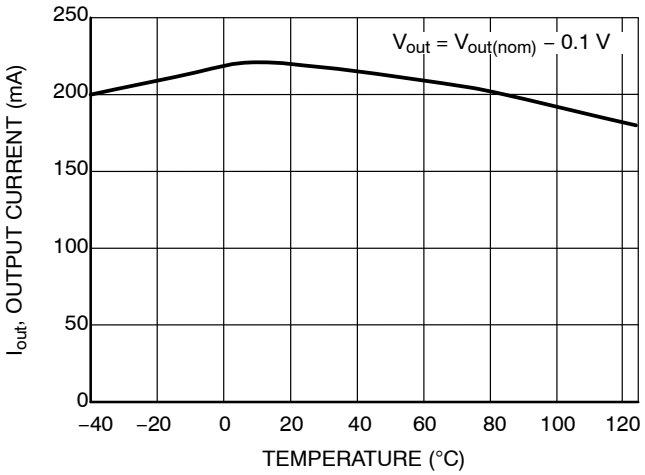


Figure 7. Output Current Limit vs. Temperature, 3.3 V

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TYPICAL CHARACTERISTICS

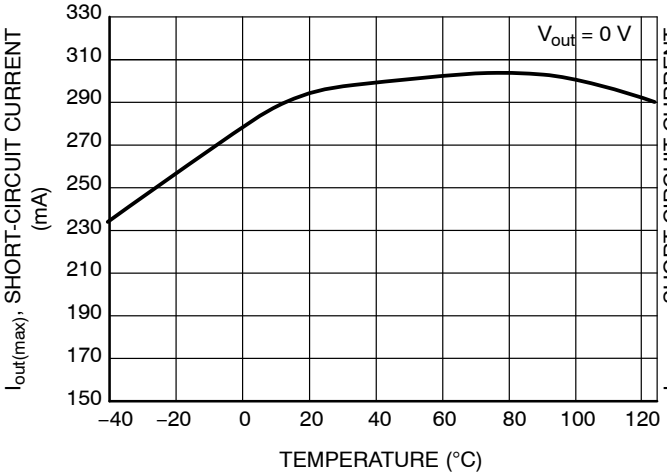


Figure 8. Short-Circuit Current Limit vs. Temperature, 1.5 V

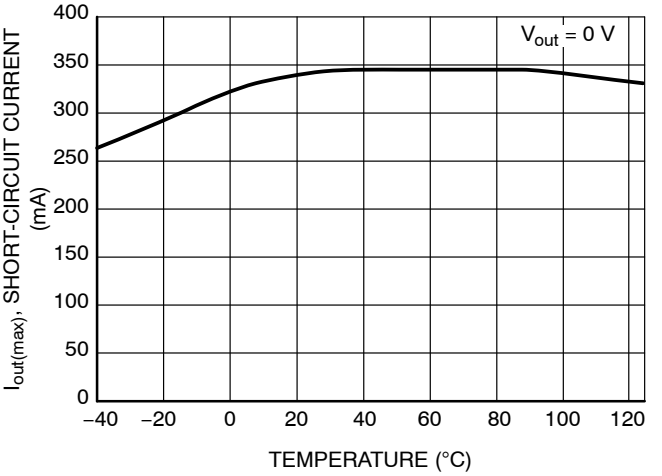


Figure 9. Short-Circuit Current Limit vs. Temperature, 3.3 V

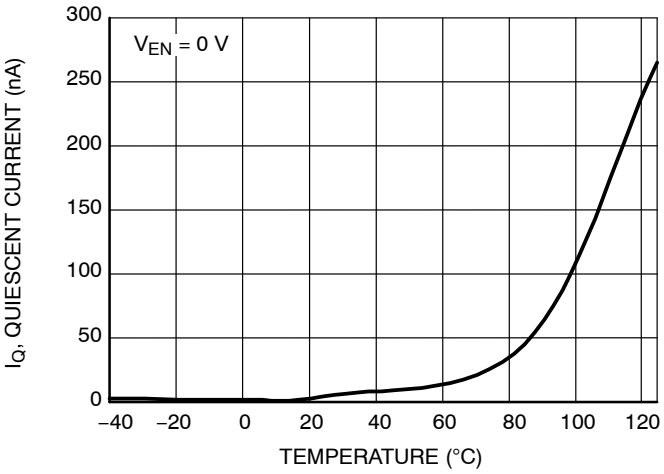


Figure 10. Quiescent Current vs. Temperature, 1.5 V

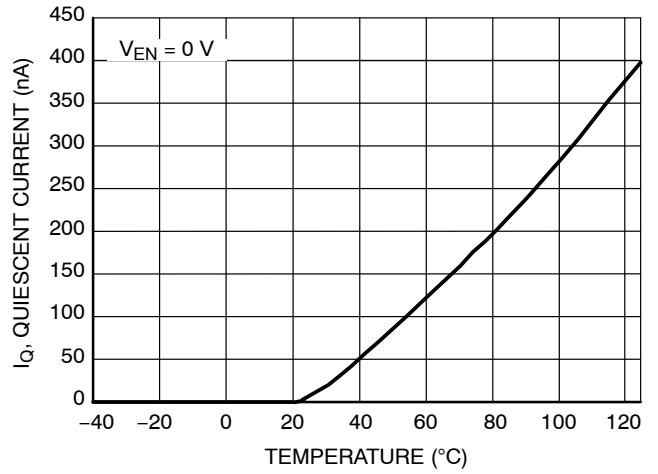


Figure 11. Quiescent Current vs. Temperature, 3.3 V

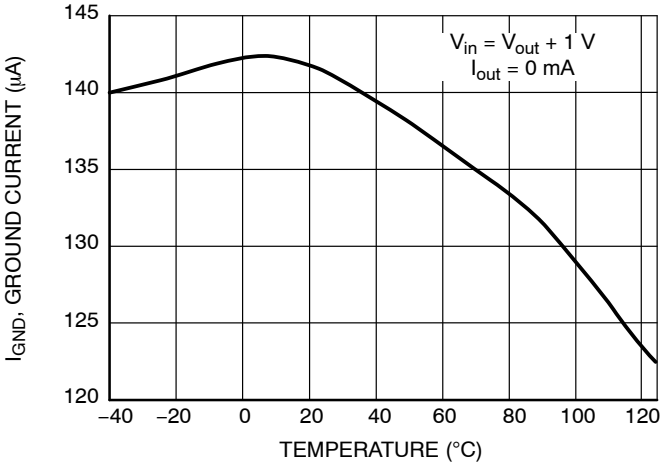


Figure 12. Ground Current vs. Temperature, 1.5 V

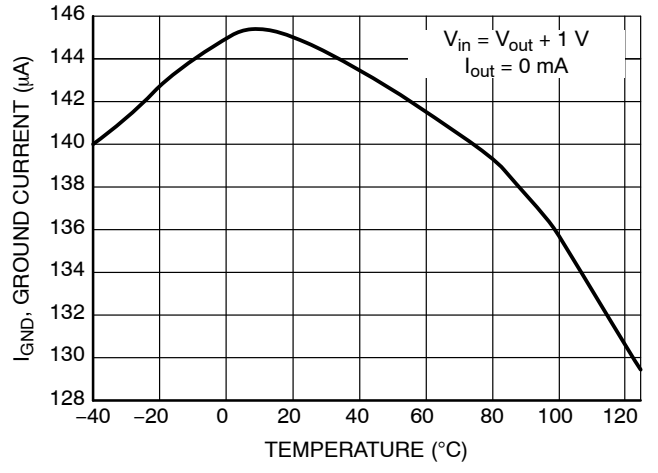


Figure 13. Ground Current vs. Temperature, 3.3 V

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TYPICAL CHARACTERISTICS

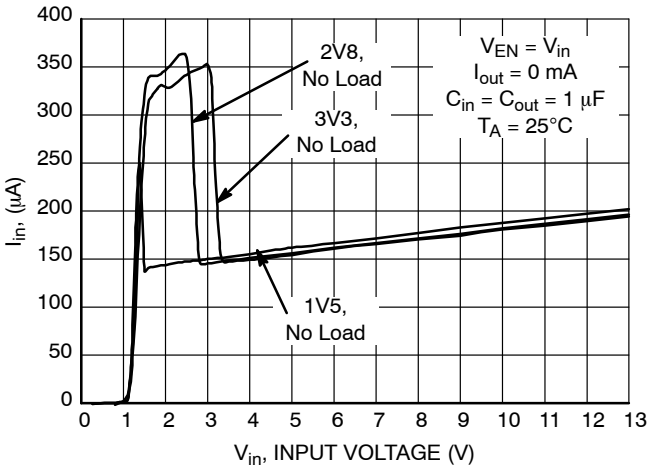


Figure 14. Quiescent Current vs. Input Voltage

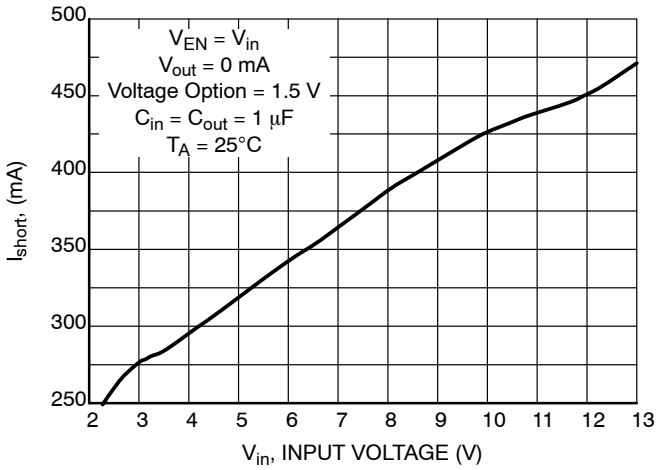


Figure 15. Output Short-Circuit Current vs. Input Voltage

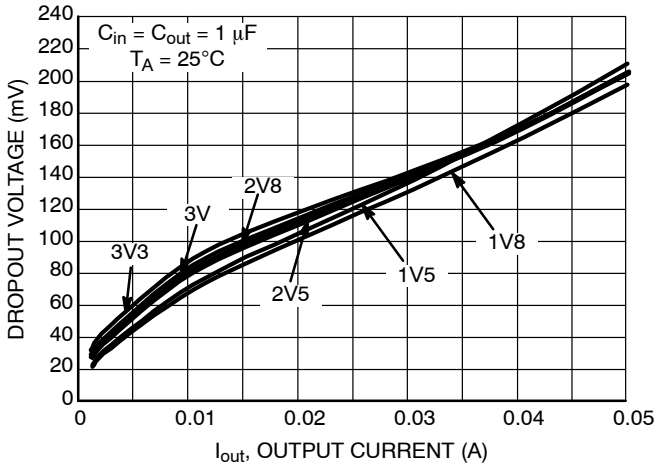


Figure 16. Dropout Voltage vs. Output Current

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TYPICAL CHARACTERISTICS

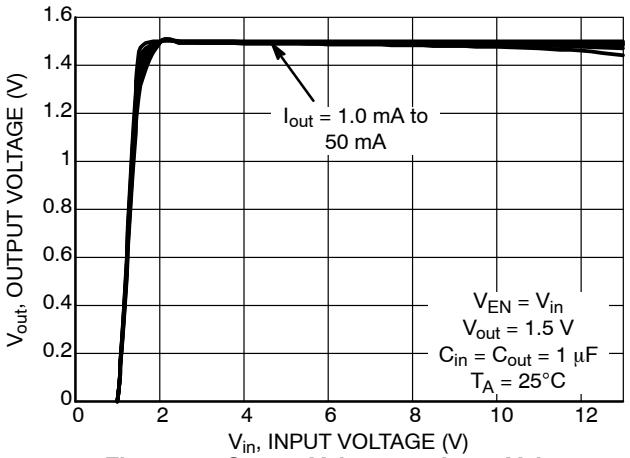


Figure 17. Output Voltage vs. Input Voltage

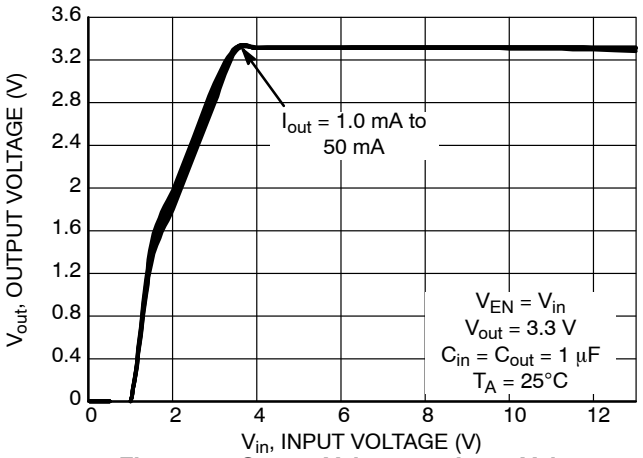


Figure 18. Output Voltage vs. Input Voltage

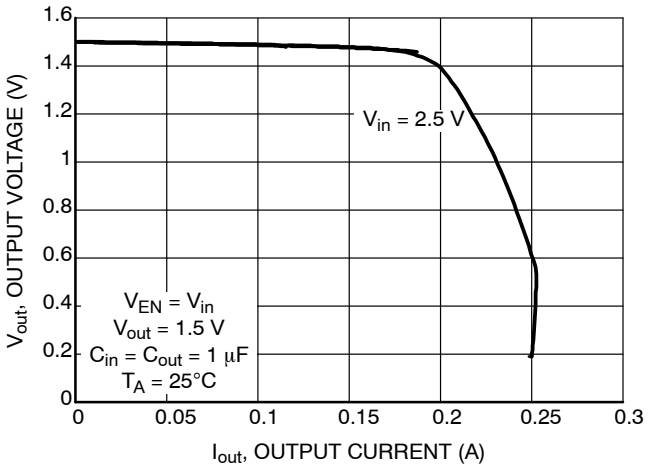


Figure 19. Output Voltage vs. Output Current

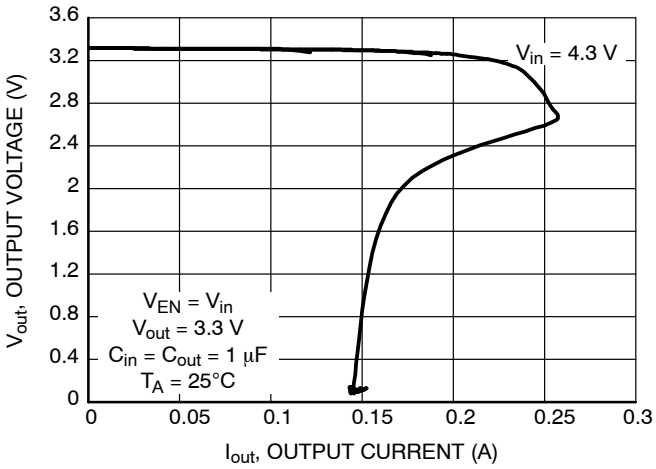


Figure 20. Output Voltage vs. Output Current

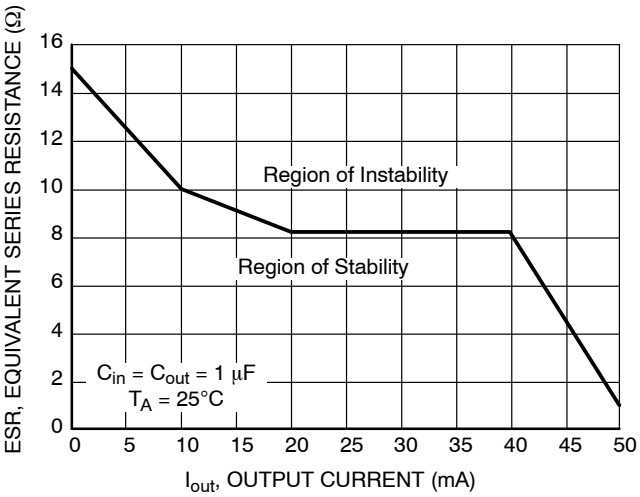


Figure 21. Equivalent Series Resistance vs. Output Current, X7R, MLCC Capacitor

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TYPICAL CHARACTERISTICS

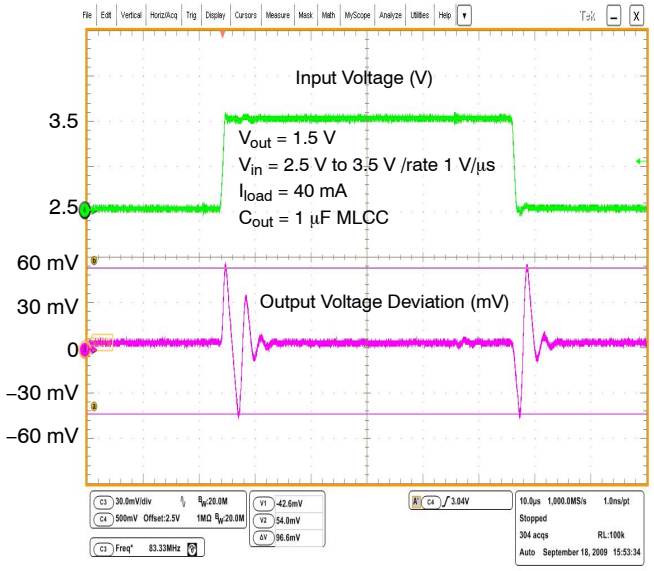


Figure 22. Line Transient Response 1.5 V/40 mA

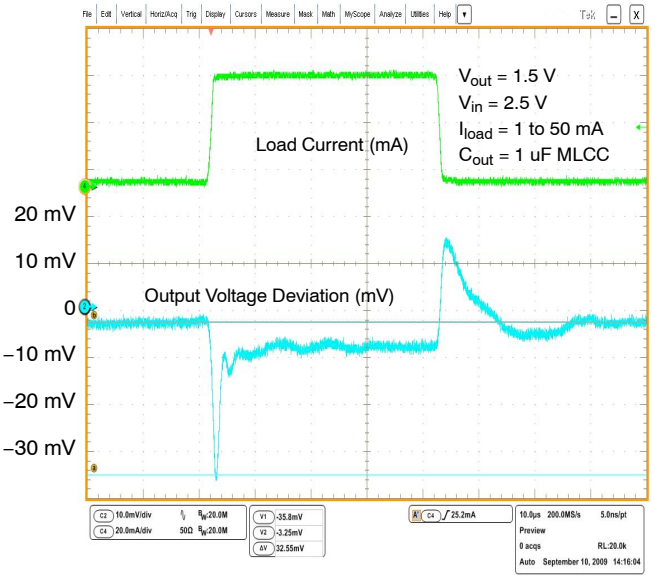


Figure 23. Load Transient Response 1.5 V

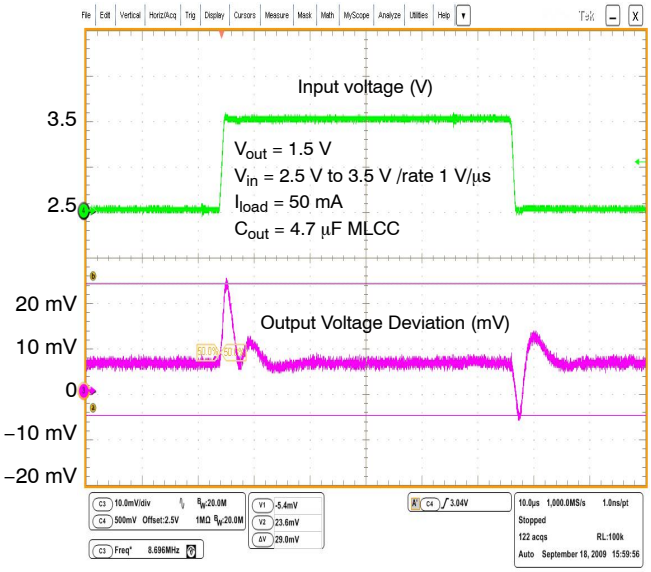


Figure 24. Line Transient Response 1.5 V/50 mA

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TYPICAL CHARACTERISTICS

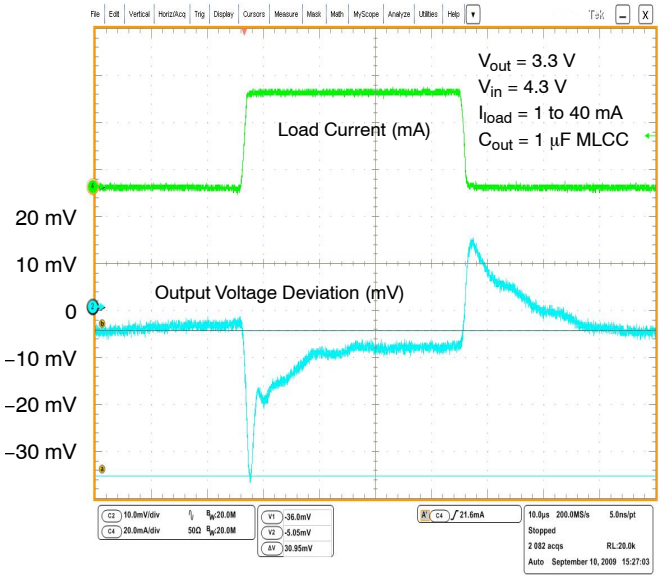


Figure 25. Load Transient Response 3.3 V

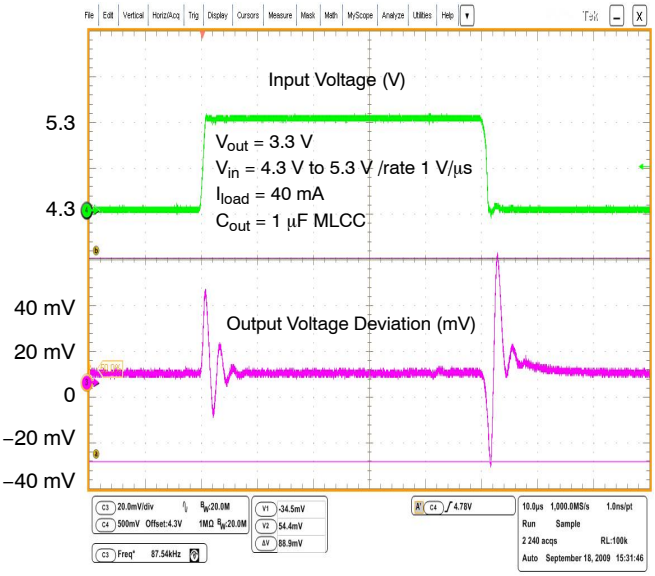


Figure 26. Line Transient Response 3.3 V/40 mA

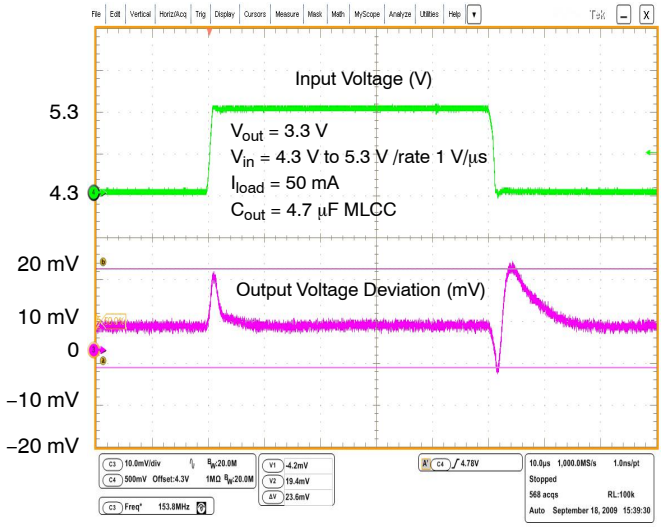


Figure 27. Line Transient Response 3.3 V/50 mA

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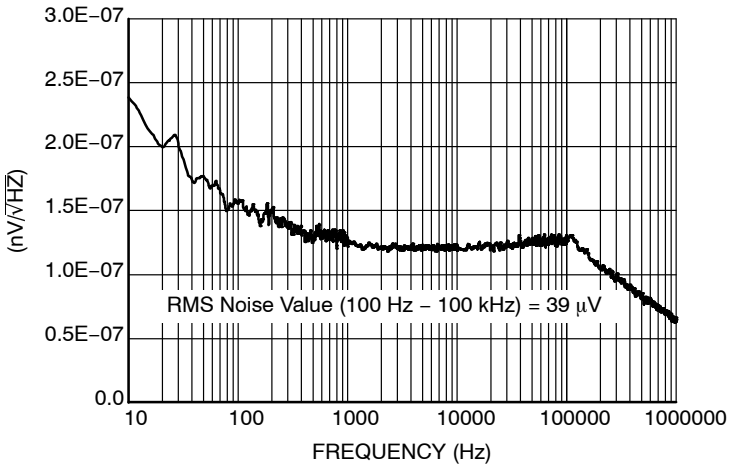


Figure 28. Output Voltage Noise
V_{out} = 1.5 V, I_{out} = 40 mA

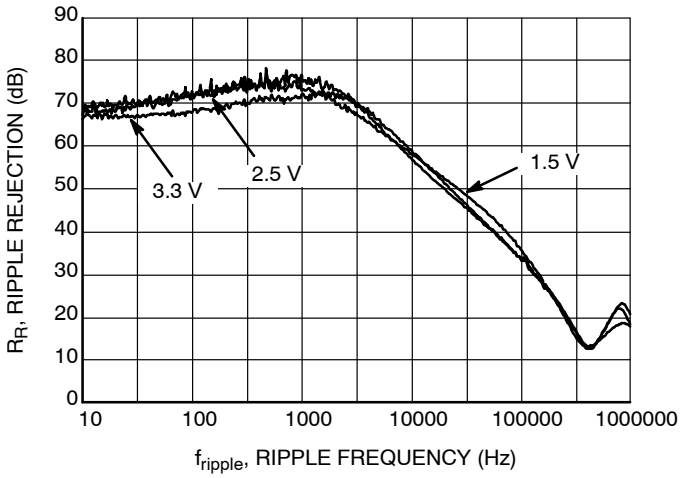


Figure 29. Ripple Rejection vs. Frequency
I_{out} = 40 mA, 0.5 V_{pp}

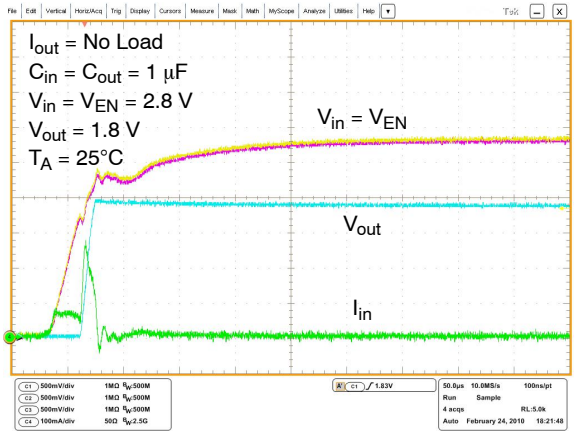


Figure 30. Startup, No Load

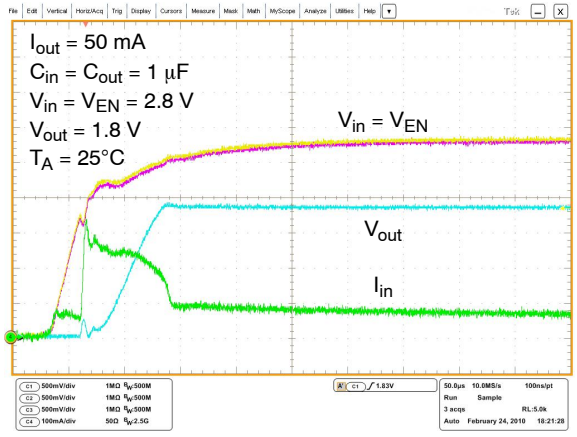


Figure 31. Startup, I_{out} = 50 mA

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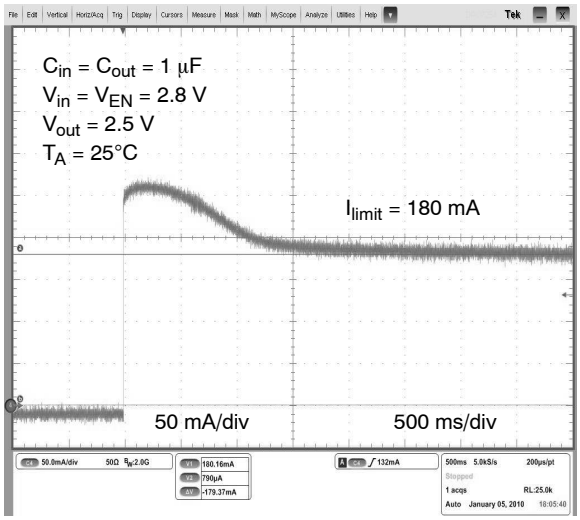


Figure 32. Hard Short-Circuit Current (by Copper Wires)

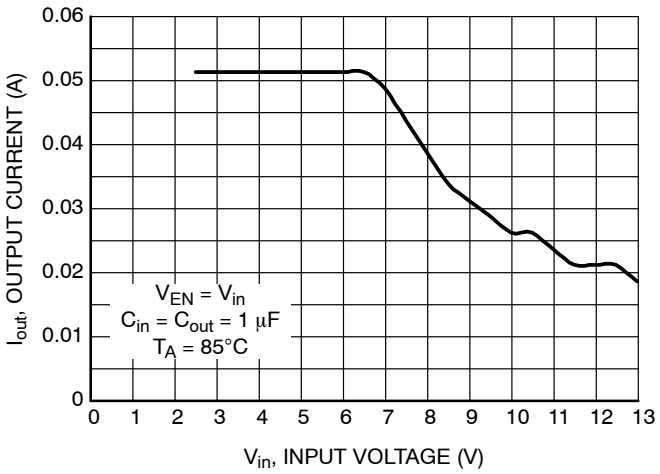


Figure 33. Measured Power Operating Area, 1.5 V, T_A = 85 °C, V_{out_drop} = max 0.1 V

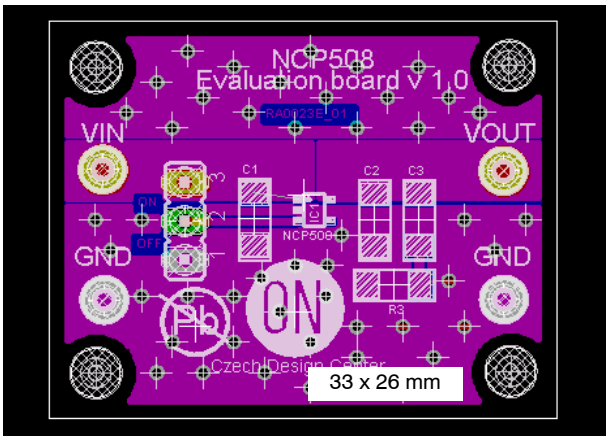


Figure 34. Evaluation Board

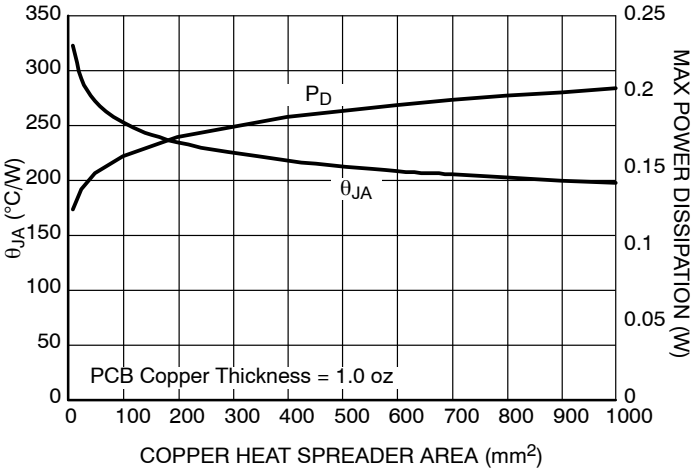


Figure 35. SC70-5 Thermal Resistance vs. Copper Heat Spreader Area

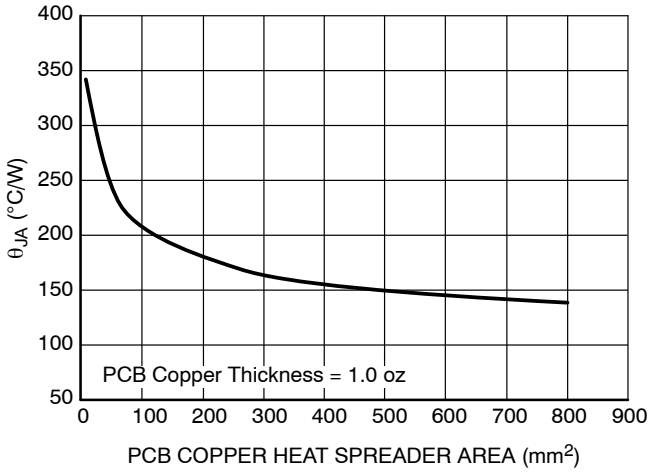


Figure 36. WDFN6 Thermal Resistance vs. Copper Heat Spreader Area

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DEFINITIONS

Load Regulation

The change in output voltage for a change in output current at a constant temperature.

Dropout Voltage

The input/output differential at which the regulator output no longer maintains regulation against further reductions in input voltage. Measured when the output drops 100 mV below its nominal. The junction temperature, load current, and minimum input supply requirements affect the dropout level.

Maximum Power Dissipation

The maximum total dissipation for which the regulator will operate within its specifications.

Quiescent Current

The quiescent current is the current which flows through the ground when the LDO operates without a load on its output: internal IC operation, bias, etc. When the LDO becomes loaded, this term is called the Ground current. It is actually the difference between the input current (measured through the LDO input pin) and the output current.

Line Regulation

The change in output voltage for a change in input voltage. The measurement is made under conditions of low dissipation or by using pulse technique such that the average chip temperature is not significantly affected.

Line Transient Response

Typical over and undershoot response when input voltage is excited with a given slope.

Thermal Protection

Internal thermal shutdown circuitry is provided to protect the integrated circuit in the event that the maximum junction temperature is exceeded. When activated at typically 125 °C, the regulator turns off. This feature is provided to prevent failures from accidental overheating.

Maximum Package Power Dissipation

The maximum power package dissipation is the power dissipation level at which the junction temperature reaches its maximum operating value, i.e. 150 °C. Depending on the ambient power dissipation and thus the maximum available output current.

APPLICATIONS INFORMATION

Typical application circuit for the NCP508 series is shown in Figure 1.

Input Decoupling (C1)

An input capacitor of at least 1.0 μF,(ceramic or tantalum) is recommended to improve the transient response of the regulator and/or if the regulator is located more than a few inches from the power source. It will also reduce the circuit's sensitivity to the input line impedance at high frequencies. The capacitor should be mounted with the shortest possible track length directly across the regular's input terminals. Higher values and lower ESR will improve the overall line transient response.

Output Decoupling (C2)

The NCP508 is a stable regulator and does not require a minimum output current. Capacitors exhibiting ESRs ranging from a few mΩ up to 3 Ω can safely be used. The minimum decoupling value is 1.0 μF and can be augmented to fulfill stringent load transient requirements. The regulator accepts ceramic chip capacitors as well as tantalum devices. Larger values improve noise rejection and load regulation transient response.

Enable Operation

The enable pin will turn on or off the regulator. The limits of threshold are covered in the electrical specification section of this data sheet. If the enable is not used then the pin should be connected to V_{in}.

Hints

Please be sure the V_{in} and GND lines are sufficiently wide. When the impedance of these lines is high, there is a chance to pick up noise or cause the regulator to malfunction.

Set external components, especially the output capacitor, as close as possible to the circuit, and make leads as short as possible.

Thermal Considerations

Internal thermal limiting circuitry is provided to protect the integrated circuit in the event that the maximum junction temperature is exceeded.

The maximum power dissipation supported by the device is dependent upon board design and layout. Mounting pad configuration on the PCB, the board material and also the ambient temperature effect the rate of temperature rise for the part. This is stating that when the NCP508 has good thermal conductivity through the PCB, the junction temperature will be relatively low with high power dissipation applications.

The maximum dissipation the package can handle is given by:

$$P_D = \frac{T_{J(max)} - T_A}{R_{\theta JA}} \qquad \text{(eq. 1)}$$

where:

- $T_{J\{max)}$ is the maximum allowable junction temperature of the die, which is 150 °C
- T_A is the ambient operating temperature
- $R_{\theta ja}$ is dependent on the surrounding PCB layout

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ORDERING INFORMATION

Device	Nominal Output Voltage	Marking	Package	Shipping†
NCP508SQ15T1G	1.5	D5A	SC-88A (Pb-Free)	3000 / Tape & Reel
NCP508SQ18T1G	1.8	D5C	SC-88A (Pb-Free)	3000 / Tape & Reel
NCP508SQ30T1G	3.0	D5F	SC-88A (Pb-Free)	3000 / Tape & Reel
NCP508SQ33T1G	3.3	D5G	SC-88A (Pb-Free)	3000 / Tape & Reel
NCP508MT15TBG	1.5	B	WDFN6 (Pb-Free)	3000 / Tape & Reel
NCP508MT25TBG	2.5	C	WDFN6 (Pb-Free)	3000 / Tape & Reel

DISCONTINUED (Note 7)

NCP508SQ25T1G	2.5	D5D	SC-88A (Pb-Free)	3000 / Tape & Reel
NCP508SQ28T1G	2.8	D5E	SC-88A (Pb-Free)	3000 / Tape & Reel
NCP508MT18TBG	1.8	A	WDFN6 (Pb-Free)	3000 / Tape & Reel
NCP508MT28TBG	2.8	D	WDFN6 (Pb-Free)	3000 / Tape & Reel
NCP508MT30TBG	3.0	E	WDFN6 (Pb-Free)	3000 / Tape & Reel
NCP508MT33TBG	3.3	F	WDFN6 (Pb-Free)	3000 / Tape & Reel

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

NOTE: Additional voltages in 100 mV steps are available upon request by contacting your **onsemi** representative.

7. **DISCONTINUED:** These devices are not available. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.

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REVISION HISTORY

Revision	Description of Changes	Date
6	Rebranded the Data Sheet to **onsemi** format. NCP508SQ25T1G, NCP508SQ28T1G, NCP508MT18TBG, NCP508MT28TBG, NCP508MT30TBG, NCP508MT33TBG OPNs Marked as Discontinued.	1/29/2026

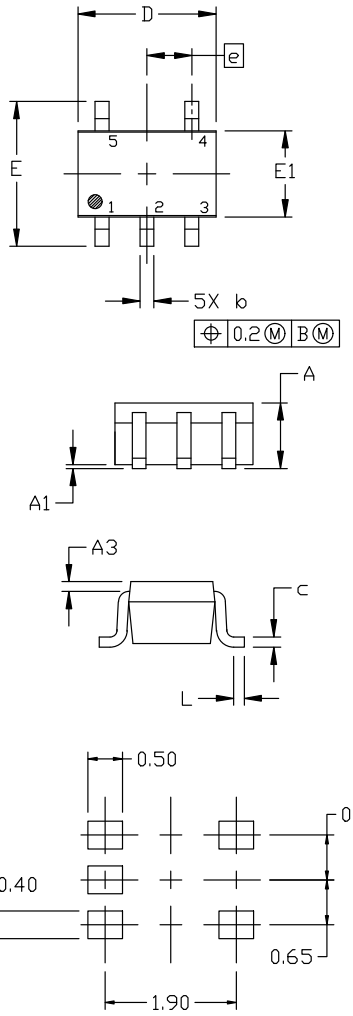
This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

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SC−88A (SC−70−5/SOT−353)
CASE 419A−02
ISSUE M

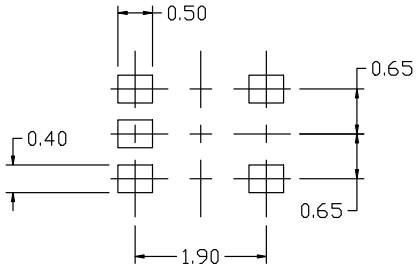
DATE 11 APR 2023

SCALE 2:1



NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: MILLIMETERS
3. 419A−01 OBSOLETE. NEW STANDARD 419A−02
4. DIMENSIONS D AND E1 DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS. MOLD FLASH, PROTRUSIONS, OR GATE BURRS SHALL NOT EXCEED 0.1016MM PER SIDE.

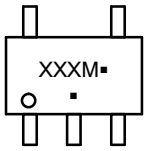
DIM	MILLIMETERS		
	MIN.	NOM.	MAX.
A	0.80	0.95	1.10
A1	---	---	0.10
A3	0.20 REF		
b	0.10	0.20	0.30
c	0.10	---	0.25
D	1.80	2.00	2.20
E	2.00	2.10	2.20
E1	1.15	1.25	1.35
e	0.65 BSC		
L	0.10	0.15	0.30



RECOMMENDED
MOUNTING FOOTPRINT

* For additional information on our Pb−Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

(Note: Microdot may be in either location)

STYLE 1:
PIN 1. BASE
2. EMITTER
3. BASE
4. COLLECTOR
5. COLLECTOR

STYLE 2:
PIN 1. ANODE
2. EMITTER
3. BASE
4. COLLECTOR
5. CATHODE

STYLE 3:
PIN 1. ANODE 1
2. N/C
3. ANODE 2
4. CATHODE 2
5. CATHODE 1

STYLE 4:
PIN 1. SOURCE 1
2. DRAIN 1/2
3. SOURCE 1
4. GATE 1
5. GATE 2

STYLE 5:
PIN 1. CATHODE
2. COMMON ANODE
3. CATHODE 2
4. CATHODE 3
5. CATHODE 4

STYLE 6:
PIN 1. EMITTER 2
2. BASE 2
3. EMITTER 1
4. COLLECTOR
5. COLLECTOR 2/BASE 1

STYLE 7:
PIN 1. BASE
2. EMITTER
3. BASE
4. COLLECTOR
5. COLLECTOR

STYLE 8:
PIN 1. CATHODE
2. COLLECTOR
3. N/C
4. BASE
5. EMITTER

STYLE 9:
PIN 1. ANODE
2. CATHODE
3. ANODE
4. ANODE
5. ANODE

Note: Please refer to datasheet for style callout. If style type is not called out in the datasheet refer to the device datasheet pinout or pin assignment.

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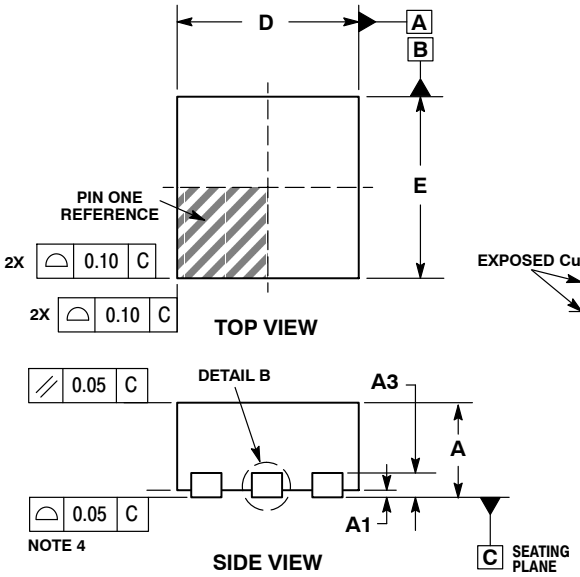
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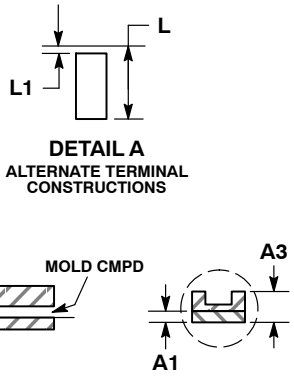
WDFN6 1.5x1.5, 0.5P
CASE 511BJ
ISSUE C

DATE 06 OCT 2015



SCALE 4:1

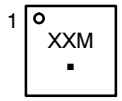




NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 1994.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. DIMENSION b APPLIES TO PLATED TERMINAL AND IS MEASURED BETWEEN 0.15 AND 0.30mm FROM TERMINAL TIP.
4. COPLANARITY APPLIES TO THE EXPOSED PAD AS WELL AS THE TERMINALS.

	MILLIMETERS	
DIM	MIN	MAX
A	0.70	0.80
A1	0.00	0.05
A3	0.20 REF	
b	0.20	0.30
D	1.50 BSC	
E	1.50 BSC	
e	0.50 BSC	
L	0.40	0.60
L1	---	0.15
L2	0.50	0.70

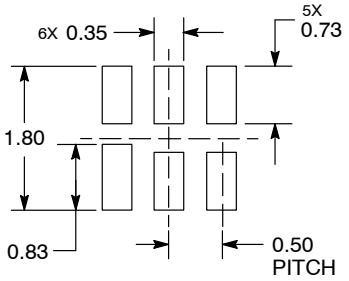
GENERIC MARKING DIAGRAM*



XX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “ ▪”, may or may not be present.

RECOMMENDED MOUNTING FOOTPRINT*



DIMENSIONS: MILLIMETERS

*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

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